Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/02/2020
PI Name:	Sharma, Anup Dutt Ph.D.		
Project Title:	Establish the Feasibility of Engineering a Human Optic-Nerve-On-A-Chip as an In Vitro Model for SANS Research		
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PI Email:	anup.d.sharma@axosim.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	504-609-3469
Organization Name:	AxoSim, Inc.		
PI Address 1:	1441 Canal St		
PI Address 2:	Ste 2016		
PI Web Page:			
City:	New Orleans	**State:**	LA
Zip Code:	70112	**Congressional District:**	2
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Moore, Michael Ph.D. (AxoSim, Inc.)		
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Task Description:	Focused Investigation Project During space travel, astronauts have to deal with long term exposure to zero gravity, galactic cosmic rays, elevated CO2, etc. The adverse effects of these challenging conditions result in a baffling physiological disorder such as Spaceflight Associated Neuro-ocular Syndrome (SANS). To understand the progression of SANS, in vitro human cells-based 3D biomimetic models capable of mimicking some of these adverse ocular symptoms is highly desired.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:

Exploring outer space is important for the long-term survival of species that have evolved over millions of years in this ever-changing cosmos. The question of whether life in its current form can survive on other planets or whether other planets already have life in some form has intrigued people for centuries, and the meteoric advances in rocket technology in the last few decades have resulted in prospects of ambitious space travel projects such as interplanetary missions. However, advances in rocket technology have not matched with our understanding of how long-term space travel affects human biology. Once in space, whether in the International Space Station (ISS) or while travel on possible future Mars missions, astronauts likely have to deal with long term exposure to zero gravity, ionizing cosmic rays, hypoxia, inflammation, etc. The adverse effects of these extremely challenging conditions on the physiological functions of the human body are largely unknown. Controlled ISS based experiments have revealed that long term space travel causes an alteration in specific genes, which impacts several important processes in the human body, including mitochondrial stress, genetic processes, connective tissue processes, and immune activity. Furthermore, it has been shown that microgravity can affect many aspects of nervous system physiology, such as vestibular and neuromuscular function, sympathetic tone and reflexes, vision, and even central processing and memory. One interesting and quite baffling physiological disorder that has been carefully documented by NASA in astronauts during and after long term spaceflights is spaceflight-associated neuro-ocular syndrome (SANS). This pathology causes unilateral and bilateral optic disc edema, globe flattening, choroidal and retinal folds, hyperopic refractive error shifts, nerve fiber layer infarcts, and thickening. The mechanism of SANS is currently a topic of great interest, and a single unifying mechanism has not yet been postulated. A possibility of multifactorial pathogenesis exists, and to study the progression, in vitro models capable of mimicking some of these adverse ocular symptoms is highly desired. The first step towards this is the development of a healthy 3D biomimetic ocular system, which mimics the development of both retinae as well as the optic nerve. In order to recreate SANS related pathophysiology in vitro, it was important to create a microphysiological system which not only begins to generate the complexity of the retina but is also capable of mimicking the optic nerve tract. Previously, we had successfully been able to create motor and sensory nerves in vitro using induced pluripotent stem cells (iPSC)-derived motor and sensory neurons using commercially available cell sources. However, currently, there is no commercially available source of retinal organoids and retinal ganglion cells. To create optic nerve-on-a-chip (NoaC), we took advantage of already published protocols. They were evaluated to determine the most effective protocol to differentiate iPSCs into retinal ganglion cells (RGCs), and perhaps more importantly, generated differentiated retinal organoids capable of dense 3D axonal outgrowth critical to the creation of a benchtop model of optic-nerve-on-a-chip. In the future, this optic NoaC model can be used for modeling diseases such as SANS and glaucoma.

Task Progress:

Original project aims/objectives
Aim 1: Differentiate retinal ganglion cells from human commercially cleared induced pluripotent stem cells (iPSCs) based on published protocols and characterize phenotypes.

Aim 2: Adapt optimal differentiation protocols for spheroid neuro-glial co-culture, based on experience with other human cell types.

Key findings

In Aim 1, we planned to differentiate human iPSCs into retinal ganglion cells (RGCs) using published protocols. We were successfully able to create human iPSC derived embryoid bodies, which were further differentiated into retinal organoids. Various stages of differentiation were characterized by the presence of CHX10 (retinal progenitor marker), BRN3A (retinal ganglion cells), and RBPMS (a selective marker of ganglion cells in the retina). Retinal organoids were found to have different levels of BRN3A expression, and approximately ~5-15% of the cells in the organoids were found to be BRN3A positive after cryosectioning and immunostaining. Interestingly, a marker of retinal progenitor cells, CHX10, was observed in retinal organoids for early (20 days) as well as late (50 days) time point. Additionally, we also compared imaging the immunostaining cryosectioned vs. whole retinal organoids. We found cryosectioned retinal organoids to be more effective in providing the details needed to observe retinal markers as well as quantification of the cells stained with those markers. At later stages of the retinal organoids culture (>40days), we also immunostained the organoids for axonal marker neurofilament-H (NF-H) and dendritic marker (MAP2) and observed the presence of axons as well as dendrites on the outer periphery of the retinal organoid. Once we realized that axons were present in these retinal organoids, we then placed the organoids in the bulb region of the nerve-on-a-chip (NoaC) platform to evaluate the growth of axons in the channel region. The bulb region resembles the retina, and the channel region resembles the optic nerve fiber track in the NoaC platform. Few of the organoids that survived during this 50 day long differentiation provided a significant axonal outgrowth in the channel region, which led us to hypothesize that creating optic-nerve-on-a-chip is feasible with the retinal differentiation protocol that we were using. Next, we also investigated ways to improve the number of retinal organoids formed during each differentiation experiment and also strategies to improve their survival once the organoids are formed. For this, we tested adding fetal bovine serum (FBS) to the cultures at various time points. We found that the addition of FBS resulted in an increased survival as well as an increased number of retinal organoids generated in each experiment. This increased number of organoids enabled us to try transferring the organoids to NoaC platform at different time points such as (DIV 20, 30, 40, and 50) to evaluate whether a timepoint resulted in enhanced axonal outgrowth. However, we also discovered that if we add FBS to the medium after transferring the organoid to NoaC, it stunts the growth of axons, and removal of FBS is necessary for maintaining the growth of axons. In summary, adding 10% FBS resulted in an increased number of spheroids formed, improved survivability, and removal of FBS resulted in higher axonal outgrowth after transferring the spheroids to the NoaC platform. After observing dense axonal outgrowth (>5mm) from the organoids in NoaC system, an electrophysiological analysis was conducted to test whether compound action potentials (CAP) can be evoked in these optic nerve constructs. We were able to record CAP from several optic nerve constructs showing that axons growing out of retinal organoids were electrically conductive and thus, can be used for nerve conduction velocity (NCV) measurement.

In summary, this in vitro model of optic nerve has molecular and electrophysiological markers of an optic nerve, which includes retinal ganglion cell markers, dense axonal outgrowths, and electrical conductivity. Because of the success with Aim1, we did not need to combine RGCs with other glial cells for creating the spheroids (proposed in Aim2). However, it will be valuable in the future to add those cells to provide additional trophic support to the growing axons.

Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal

The optic nerve is mostly composed of retinal ganglion cells (RGCs), the cells which are most affected due to the changes in intraocular pressure, thus resulting in degeneration of axons and permanent damage to the optic nerve.

	Spaceflight associated neuro-ocular syndrome (SANS) is one such physiological disorder that has been carefully documented by NASA in astronauts during and after long term spaceflights. One mechanism proposed for SANS is that changes in intracranial pressure may impact both the ocular structure as well as function. In order to recreate SANS related pathophysiology in vitro, it is essential to create a microphysiological system (MPS), which not only begins to generate the complexity of the retina but is also capable of mimicking the optic nerve tract. For true relevance, metrics capturing changes to the structure and function of the retina and optic nerve must be evaluated under changing pressure conditions. In this study, we were successfully able to replicate key features of an optic nerve using a human cell-based MPS taking advantage of the Nerve-on-a-Chip (NoaC) platform and previously published retinal differentiation protocols. This research helped us in determining the most effective protocol to differentiate iPSCs into RGCs, and perhaps more importantly, generate differentiated RGCs capable of dense 3D axonal outgrowth critical to the creation of a benchtop model of optic-nerve-on-a-chip. In the future, this optic NoaC model can be used for modeling diseases such as SANS and glaucoma.
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